

Midterm 2: Terminal Trace Diagnostics

Data Plane (Ch 4), Control Plane (Ch 5), and ARP Transition

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Midterm 2 Preparation Strategy

- **The Exam:** 60 minutes, highly practical multiple-choice questions.
- **The Focus:** You will be given simulated terminal outputs and asked to trace what is happening inside the Network Layer (Data & Control planes).
- **Core Commands to Know:**
 - ▶ `ip route show` (Data Plane: Forwarding Tables)
 - ▶ `dhclient / journalctl` (Data Plane: DHCP Acquisition)
 - ▶ `conntrack -L` (Data Plane: NAT Tracking)
 - ▶ `ping & traceroute` (Control Plane: ICMP Error Generation)
 - ▶ `arp -n` (Link Layer Bridge: MAC Address Resolution)

Data Plane Command: `ip route show`

Concept: How the router decides where to send a packet (Forwarding).

```
$ ip route show
default via 192.168.1.1 dev eth0
10.1.1.0/24 dev eth1 proto kernel scope link src 10.1.1.1
10.1.1.128/26 dev eth2 proto kernel scope link src 10.1.1.129
10.1.1.192/28 dev eth3 proto kernel scope link src 10.1.1.193
```

Trace Scenario: A packet arrives destined for 10.1.1.200. Which interface (eth) does it leave from?

- **Step 1: Find the matches.** .200 falls inside the /24 (0-255) and the /28 (192-207). It does *not* fit the /26 (128-191).
- **Step 2: Apply the Rule.** Routers use **Longest Prefix Match**. They always choose the most specific subnet mask.
- **Result:** /28 is longer (more specific) than /24. The packet is forwarded out eth3.

Data Plane Command: `dhclient -v`

Concept: The D.O.R.A process for acquiring an IP address.

```
$ sudo dhclient -v eth0
```

```
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 3
```

```
DHCPOFFER of 192.168.1.100 from 192.168.1.1
```

```
DHCPREQUEST for 192.168.1.100 on eth0 to 255.255.255.255 port 67
```

```
DHCPACK from 192.168.1.1
```

Trace Scenario: Look at the DHCPREQUEST line. What is the Source and Destination IP in the packet header at that exact moment?

- **Source IP:** 0.0.0.0. The client was *offered* .100, but it cannot legally use it until the ACK arrives.
- **Destination IP:** 255.255.255.255. The client broadcasts its acceptance so that *all* other DHCP servers on the subnet know to cancel their pending offers.

Data Plane Command: `conntrack -L`

Concept: Network Address Translation (NAT) table tracking.

```
$ sudo conntrack -L
```

```
tcp 6 ESTABLISHED src=10.1.1.55 dst=142.250.113.1 sport=50222 dport=
      src=142.250.113.1 dst=198.51.100.4 sport=443 dport=
```

Trace Scenario: The external web server (142...) is replying to your laptop (10.1.1.55). Before the packet reaches your router, what is its Destination IP?

- **The Rule:** Private IPs (10.x.x.x) cannot exist on the public internet. The web server does not know your laptop exists.
- **The Destination:** The server sends the reply to the router's public WAN IP: 198.51.100.4.
- **The Translation:** The router's hardware receives it, checks this table, and swaps the destination back to 10.1.1.55 before sending it to the LAN.

Control Plane Command: ping

Concept: ICMP Error Generation in the Control Plane.

```
$ ping 10.5.5.50
PING 10.5.5.50 (10.5.5.50) 56(84) bytes of data.
From 192.168.2.1 icmp_seq=1 Destination Net Unreachable
From 192.168.2.1 icmp_seq=2 Destination Net Unreachable
```

Trace Scenario: What is the exact root cause of this failure? Is the server offline?

- **Read the Source:** The error is coming from an intermediate gateway (192.168.2.1), *not* the target server.
- **The Meaning:** "Destination Net Unreachable" is a routing failure. The router checked its forwarding table and had no path to the 10.5.5.0 network.
- **Contrast:** If the server was simply offline, the router would successfully forward it, and the terminal would say "*Request Timed Out*".

Control Plane Command: `traceroute -n`

Concept: Using TTL to map routing paths via ICMP.

```
$ traceroute -n 142.250.113.1
```

```
1  192.168.0.1      1.2 ms
2  10.14.22.1       15.4 ms
3  * * *
4  172.217.14.2     24.3 ms
5  142.250.113.1    25.3 ms
```

Trace Scenario: Why are there asterisks at Hop 3? Did the network drop the packet due to congestion?

- **Data Plane is Fine:** The packet did *not* drop. We know this because Hop 4 and Hop 5 successfully replied!
- **Control Plane Block:** Traceroute relies on routers sending an ICMP "Time Exceeded" error when TTL hits 0. The administrator at Hop 3 configured their firewall to silently drop outbound ICMP errors.

Link Layer Command: `arp -n`

Concept: Bridging Logical IPs to Physical MACs. **Setup:** Your IP is 192.168.10.65/26. Your gateway is 192.168.10.126.

```
$ arp -n
```

Address	HWaddress	Flags
192.168.10.66	00:11:22:33:44:55	C
192.168.10.126	aa:bb:cc:dd:ee:01	C

Trace Scenario: You execute `ping 192.168.10.130`. Does the computer broadcast an ARP request? If not, what is the Destination MAC?

- **Step 1 (Subnetting):** A /26 mask means your local network ranges from .64 to .127.
- **Step 2 (Local vs Remote):** The target .130 is outside your subnet. It is on a *remote* network.
- **Step 3 (The Gateway):** Because it is remote, you do NOT broadcast ARP. You encapsulate the packet inside a frame destined for the Default Gateway's MAC: aa:bb:cc:dd:ee:01.

Managing the Cache: Flushing ARP

Concept: Forcing the network stack to re-learn physical addresses.

Modern Linux command to clear the ARP Cache:

```
$ sudo ip neigh flush all
```

Verifying the cache is empty:

```
$ arp -n
```

(No entries found)

Trace Scenario: Immediately after running this, you ping a device on your *local* subnet. What happens?

- Because the cache is empty, the host must pause the IP datagram.
- It generates an ARP Request to discover the physical address.
- It sends an Ethernet frame to the Broadcast MAC: `ff:ff:ff:ff:ff:ff`.
- Every device on the switch receives and processes this frame.

Read the Terminal, Not Your Assumptions!

- Before answering any ARP question on the exam, calculate the subnet to see if the target is local or remote.
- If a ping fails, check the IP address that is reporting the error.
- Remember the separation: A ping timeout means the packet reached the end. A "Destination Unreachable" means a router's forwarding table failed.